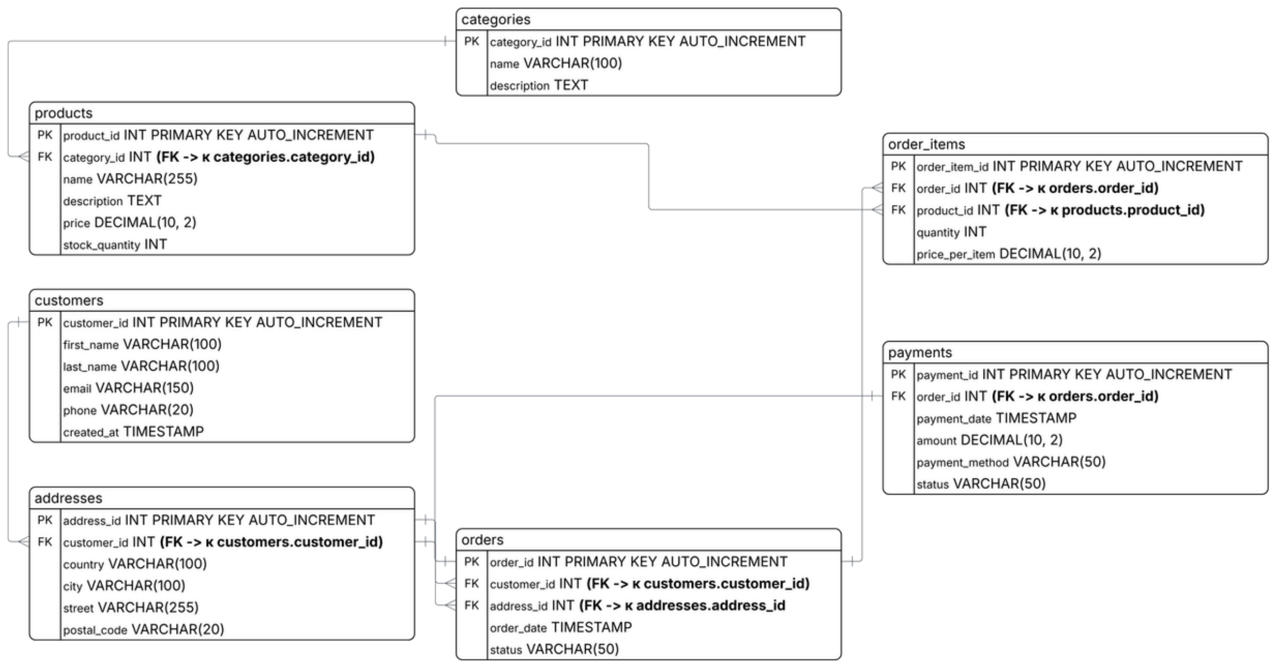

Entities & Attributes

1. customer: customer_id, first_name, last_name, email, phone, created_at
2. address: address_id, customer_id, country, city, street, postal_code
3. category: category_id, name, description
4. product: product_id, category_id, name, sku, price, discount, final_price, stock_quantity
5. order: order_id, customer_id, shipping_address_id, order_date, status
6. order_item: order_item_id, order_id, product_id, quantity, price_per_item
7. payment: payment_id, order_id, payment_date, amount, payment_method, status

Relationships

- customer ➔ address (1:N): A customer manages many saved addresses; an address belongs to exactly one customer.
- customer ➔ order (1:N): A customer places many orders; an order belongs to exactly one customer.
- address ➔ order (1:N): An address receives many orders; an order is shipped to exactly one address.
- category ➔ product (1:N): A category contains many products; a product belongs to exactly one category.
- order ➔ product (N:M): An order contains many products, and a product can be part of many orders. (Resolved via order_item).
- order ➔ order_item (1:N): An order contains many order items; an order item belongs to exactly one order.
- product ➔ order_item (1:N): A product appears in many order items; an order item references exactly one product.
- order ➔ payment (1:N): An order can have many payment attempts (e.g., if one fails); a payment belongs to exactly one order.

Logical Model Table

TABLE	COLUMN	DATA TYPE	NULLABLE	KEY	CONSTRAINTS & NOTES
customer	customer_id	INT AUTO_INCREMENT	no	PK	Primary Key
	first_name	VARCHAR(100)	no		Customer given name
	last_name	VARCHAR(100)	no		Customer surname
	email	VARCHAR(150)	no	UNIQUE	Condition 4 Natural Key Constraint
	phone	VARCHAR(20)	yes		Contact phone number
	created_at	TIMESTAMP	no		DEFAULT CURRENT_TIMESTAMP
address	address_id	INT AUTO_INCREMENT	no	PK	Primary Key
	customer_id	INT	no	FK	References customer.customer_id
	country	VARCHAR(100)	no		Destination country
	city	VARCHAR(100)	no		Destination city
	street	VARCHAR(255)	no		Condition 5 NOT NULL on non-trivial column
	postal_code	VARCHAR(20)	no		Postal / ZIP code
category	category_id	INT AUTO_INCREMENT	no	PK	Primary Key
	name	VARCHAR(100)	no		Category name (e.g. Electronics)
	description	TEXT	yes		Detailed description
product	product_id		no	PK	Primary Key
		INT AUTO_INCREMENT			
	category_id	INT	yes	FK	References category.category_id
	name	VARCHAR(255)	no		Product catalog name
	sku	VARCHAR(50)	no	UNIQUE	Stock Keeping Unit (Natural Key)
	price	DECIMAL(10,2)	no		Base price
	discount	DECIMAL(10,2)	no		DEFAULT 0.00 (Promotional reduction)
	final_price	DECIMAL(10,2)	no		GENERATED ALWAYS AS (price - discount) STORED
	stock_quantity	INT	no		Condition 2 CHECK (stock_quantity >= 0)

orders	order_id	INT AUTO_INCREMENT	no	PK	Primary Key
	customer_id	INT	no	FK	References customer.customer_id
	shipping_address_id	INT	yes	FK	References address.address_id
	order_date	TIMESTAMP	no		Condition 1 CHECK (order_date > '2026-01-01')
	status	VARCHAR(50)	no		Condition 3 CHECK status options, DEFAULT 'Pending'
order_item	order_item_id	INT AUTO_INCREMENT	no	PK	Junction Table (Many-to-Many resolver)
	order_id	INT	no	FK	References orders.order_id
	product_id	INT	no	FK	References product.product_id
	quantity	INT	no		CHECK (quantity > 0)
	price_per_item	DECIMAL(10,2)	no		Historical checkout price capture
payment	payment_id	INT AUTO_INCREMENT	no	PK	Primary Key
	order_id	INT	no	FK	References orders.order_id
	payment_date	TIMESTAMP	no		DEFAULT CURRENT_TIMESTAMP
	amount	DECIMAL(10,2)	no		Processed transactional monetary value
	payment_method	VARCHAR(50)	no		Credit Card, PayPal, Cash
	status	VARCHAR(50)	no		Transaction status (e.g., Completed)

Checking

- Primary Keys & Types: Every table has a single-column INT AUTO_INCREMENT PRIMARY KEY. Money uses DECIMAL(10,2), and dates use TIMESTAMP.
- Many-to-Many Resolution: The relationship between orders and products is resolved via the order_item junction table using two foreign keys (order_id, product_id).

The 5 Required Constraints & Rules:

- CHECK (Date): orders.order_date > '2026-01-01 00:00:00' (Prevents invalid historical dates).
- CHECK (Non-negative): products.stock_quantity >= 0 (Inventory cannot drop below zero).
- CHECK (Options): orders.status IN ('Pending', 'Processing', 'Shipped', 'Delivered', 'Cancelled').
- UNIQUE (Natural Key): customers.email UNIQUE (No duplicate accounts).
- NOT NULL (Non-trivial): addresses.street NOT NULL (Cannot ship an order without a street name).
- DEFAULT: orders.status DEFAULT 'Pending' / products.discount DEFAULT 0.00.
- GENERATED Column: products.final_price DECIMAL(10,2) GENERATED ALWAYS AS (price - discount) STORED.

Referential Actions (ON DELETE):

- CASCADE: Used on address.customer_id and order_item.order_id (If the parent profile/order is deleted, its personal addresses or item lines are wiped automatically).
- RESTRICT: Used on orders.customer_id and order_item.product_id (Blocks deletion of a customer or product if they have financial history/past orders).
- SET NULL: Used on orders.shipping_address_id and product.category_id (If an address or category is deleted, historical logs remain intact with a NULL reference).

3NF Sanity Check

- 1NF (Atomicity): Passed. Every cell holds a single, indivisible scalar value. No comma-separated lists or arrays.
- 2NF (No Partial Dependency): Passed. All tables use single-column primary keys, meaning non-key columns cannot depend on only a "part" of a key.
- 3NF (No Transitive Dependency): Passed. Non-key columns depend directly on the primary key, not through other non-key columns.